Quantum Field Theory II   Homework

2.5

We define $(i\not\partial - m)\psi = \int_y i S_F^{-1}(x-y)\,\psi(y)$

$$\Rightarrow \quad i S_F^{-1}(x-y) = (-i\not\partial_y - m)\,\delta^{(4)}(x-y)$$

The fermion propagator is the Greens function of the Dirac operator:

$$(i\not\partial_x - m)\,S_F(x-y) \equiv i\,\delta(x-y)$$

This means $\int_z S_F^{-1}(x-z)\,S_F(z-y) = \delta(x-y)$ must be true for S_F^{-1} to be inverse of S_F

Let's show this for $S_F(z-y) = \int \frac{d^4q}{(2\pi)^4}\,\frac{i(\not q+m)}{q^2-m^2+i\varepsilon}\,e^{-iq\cdot(z-y)}$

$$\int_z S_F^{-1}(x-z)\,S_F(z-y) = -i\int_z (-i\not\partial_z - m)\,\delta^{(4)}(x-z)\int\frac{d^4q}{(2\pi)^4}\,\frac{i(\not q+m)}{q^2-m^2+i\varepsilon}\,e^{-iq\cdot(z-y)}$$

$$= -i\int_z\int\frac{d^4q}{(2\pi)^4}\,\frac{i(\not q+m)}{q^2-m^2+i\varepsilon}\,(-i\not\partial_z - m)\,\delta^{(4)}(x-z)\,e^{-iq\cdot(z-y)}$$

$$\int_z \left(+i\not\partial_z\, e^{-iq(z-y)}\right)\delta^{(4)}(x-z) = \int_z\left[(-i\not\partial_z)\,\delta^{(4)}(x-z)\right]e^{-iq(z-y)} + \underbrace{\text{Boundary terms}}_{=0}$$

Same integration by part as in the hints

$$= -i\int_z\int\frac{d^4q}{(2\pi)^4}\,\frac{i(\not q+m)}{q^2-m^2+i\varepsilon}\,(i\not\partial_z - m)\,e^{-iq(z-y)}\,\delta^{(4)}(x-z)$$

Now we integrate $\int_z$

$$= \int\frac{d^4q}{(2\pi)^4}\,\frac{(\not q+m)}{q^2-m^2+i\varepsilon}\,(i\not\partial_x - m)\,e^{-iq(x-y)}$$

$$\not\partial_x\, e^{-iq(x-y)} = \gamma^\mu\partial_{\mu x}\, e^{-iq(x-y)} = \gamma^\mu(-iq_\mu)\,e^{-iq(x-y)} = -i\not q\, e^{-iq(x-y)}$$

$$= \int\frac{d^4q}{(2\pi)^4}\,\frac{(\not q+m)}{q^2-m^2+i\varepsilon}\,(i(-i)\not q - m)\,e^{-iq(x-y)}$$

$$= \int\frac{d^4q}{(2\pi)^4}\,\frac{(\not q+m)(\not q-m)}{q^2-m^2+i\varepsilon}\,e^{-iq(x-y)}$$

$$\not q\cdot\not q = q^2 \quad\Rightarrow\quad (\not q+m)(\not q-m) = q^2-m^2$$

The $i\varepsilon$ prescription is there to calculate integral such that it doesn't diverge but here we have Numerator = Denominator without $i\varepsilon$, so we don't need $i\varepsilon \to \varepsilon = 0$

$$\Rightarrow \quad = \int\frac{d^4q}{(2\pi)^4}\,\frac{q^2-m^2}{q^2-m^2}\,e^{-iq(x-y)} = \int\frac{d^4q}{(2\pi)^4}\,e^{-iq(x-y)}$$

This integral is just a dirac delta function

$$\Rightarrow \quad = \delta^{(4)}(x-y) \quad\Rightarrow\quad \int_z S_F^{-1}(x-z)\,S_F(z-y) = \delta^{(4)}(x-y) \quad \square$$

we can drop (4) from $\delta^{(4)}(x-y)$ since x and y are 4-vectors

We have shown that $\int_z S_F^{-1}(x-z)\, S_F(z-y) = \delta(x-y)$

for $S_F(x-y) = \int \frac{d^4q}{(2\pi)^4} \frac{i(\not q+m)}{q^2-m^2+i\epsilon}\, e^{-iq(x-y)}$ and $iS_F^{-1}(x-y) = (-i\not\partial_y - m)\,\delta^{(4)}(x-y)$

As such $S_F^{-1}(x-y)$ that is is defined as $iS_F^{-1}(x-y) = (-i\not\partial_y - m)\,\delta^{(4)}(x-y)$

is the inverse function of the fermion propagator $S_F(x-y) = \int \frac{d^4q}{(2\pi)^4} \frac{i(\not q+m)}{q^2-m^2+i\epsilon}\, e^{-iq(x-y)}$

Exercise 2.6

Derive the generating functional for the free Dirac field,

$$Z_0[\bar\eta,\eta] = Z_0[0,0]\exp\left[-\int d^4x\, d^4y\; \bar\eta(x)\, S_F(x-y)\, \eta(y)\right]$$

The generating functional of the free Dirac field is

$$Z_0[\bar\eta,\eta] = \int D\Psi\, D\bar\Psi\, \exp\left[i\int_x \bar\Psi(i\not\partial - m)\Psi + \bar\eta\Psi + \bar\Psi\eta\right]$$

We shift the fields by:

$$\Psi'(x) = -\left[\Psi(x) - i\int_y S_F(x-y)\,\eta(y)\right]$$

This implies $\bar\Psi'(x) = -\left[\Psi'^{\dagger}(x)\gamma^0\right] = -\left(\bar\Psi(x) + i\left[\int_y \eta^\dagger(y)\, S_F^\dagger(x-y)\right]\gamma^0\right)$

$$S_F(x-y) = i\int \frac{d^4k}{(2\pi)^4} \frac{(\not k+m)}{k^2-m^2+i\epsilon}\, e^{-ik(x-y)} \Rightarrow S_F^\dagger(x-y) = -i\int \frac{d^4k}{(2\pi)^4} \frac{\left(k_\mu(\gamma^\mu)^\dagger + \gamma^0 m \gamma^0\right)}{k^2-m^2-i\epsilon}\, e^{-ik(y-x)}$$

we use $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$

$$\Rightarrow S_F^\dagger(x-y) = -i\int \frac{d^4k}{(2\pi)^4} \frac{\gamma^0(\not k+m)\gamma^0}{k^2-m^2-i\epsilon}\, e^{-ik(y-x)} = -\gamma^0 S_F(y-x)\gamma^0$$

$$\Rightarrow \bar\Psi'(x) = -\left[\bar\Psi(x) - i\int_y \bar\eta(y)\, S_F(y-x)\right]$$

$$\Rightarrow \bar\eta\Psi(x) \to -\left[\bar\eta(x)\Psi(x) - \int_y \bar\eta(x)\, S_F(x-y)\,\eta(y)\right]$$

$$\bar\Psi(x)\eta(x) \to -\left[\bar\Psi(x)\eta(x) - \int_y \bar\eta(y)\, S_F(y-x)\,\eta(x)\right]$$

Before we apply the shif we modify the functional. We use the relation

$(i\not\partial - m)\Psi(x) = \int_y iS_F^{-1}(x-y)\,\Psi(y)$ to get

$$Z_0[\bar\eta,\eta] = \int D\Psi D\bar\Psi \exp\left[i\int_x\int_y \bar\Psi(x)\, i S_F^{-1}(x-y)\Psi(y) + i\int_x \bar\eta\Psi + \bar\Psi\eta\right]$$

A shift of $\Psi(x) \to \Psi'(x)$ and $\bar\Psi(x) \to \bar\Psi'(x)$ is linear so

$$D\Psi D\bar\Psi \to D\Psi' D\bar\Psi' = -D\Psi(-D\bar\Psi) = D\Psi D\bar\Psi$$

So after the shift we have

$$Z_0[\bar\eta,\eta] = \int D\Psi D\bar\Psi \exp\left[i\int_x\int_y \bar\Psi'(x)\, i S_F^{-1}(x-y)\Psi'(y) + i\int_x \bar\eta\Psi' + \bar\Psi'\eta\right]$$

Let's evaluate the integrals in the exponential separately

$$\int_x \bar\eta\Psi' = \int_x -\left[\bar\eta(x)\Psi(x) - i\int_y \bar\eta(x) S_F(x-y)\eta(y)\right]$$

$$= -\int_x \bar\eta\Psi + i\int_x\int_y \bar\eta(x) S_F(x-y)\eta(y)$$

$$\int_x \bar\Psi'\eta = \int_x -\left[\bar\Psi(x)\eta(x) - i\int_y \bar\eta(y) S_F(y-x)\eta(x)\right]$$

$$= -\int_x \bar\Psi\eta + i\int_x\int_y \bar\eta(y) S_F(y-x)\eta(x)$$

$$= -\int_x \bar\Psi\eta + i\int_x\int_y \bar\eta(x) S_F(x-y)\eta(y)$$

By renaming of variables the integral on right becomes: $\int_y\int_x \bar\eta(x) S_F(x-y)\eta(y)$

Integration order doesn't matter so it's: $\int_x\int_y \bar\eta(x) S_F(x-y)\eta(y)$

$$\Rightarrow \quad i\int_x \bar\eta\Psi' + \bar\Psi'\eta = -i\int_x \bar\eta\Psi + \bar\Psi\eta + 2i\int_x\int_y \bar\eta(x) i S_F(x-y)\eta(y)$$

Now we calculate $\int_x\int_y \bar\Psi'(x) i S_F^{-1}(x-y)\Psi'(y)$

$$= \int_x\int_y -\left[\bar\Psi(x) - i\int_z \bar\eta(z) S_F(z-x)\right] i S_F^{-1}(x-y)\left(-\left[\Psi(y) - i\int_{z'} S_F(y-z')\eta(z')\right]\right)$$

$$= +\int_x\int_y \bar\Psi(x) i S_F^{-1}(x-y)\Psi(y) - i\int_x\int_y\int_z \bar\eta(z) S_F(z-x) i S_F^{-1}(x-y)\Psi(y)$$

We first integrate this over x to get delta function using $\int_y S_F^{-1}(x-y) S_F(y-z) = \delta$

$$- i\int_x\int_y\int_{z'} \bar\Psi(x) i S_F^{-1}(x-y) S_F(y-z')\eta(z') + i^2\int_x\int_y\int_z\int_{z'} \bar\eta(z) S_F(z-x) i S_F^{-1}(x-y) S_F(y-z')\eta(z')$$

↑ First integrate over y ↑ integrate over y

$$\Rightarrow \; = \int_x\int_y \bar\Psi(x) i S_F^{-1}(x-y)\Psi(y) - i^2\int_y\int_z \bar\eta(z)\,\delta(z-y)\,\Psi(y) - i^2\int_x\int_{z'} \bar\Psi(x)\,\delta(x-z')\,\eta(z')$$
$$+ i^2 i \int_x\int_z\int_{z'} \bar\eta(z)\, S_F(z-x)\,\delta(x-z')\,\eta(z')$$

Now we integrate over the delta-functions

$$\Rightarrow \; = \int_x\int_y \bar\Psi(x) i S_F^{-1}(x-y)\Psi(y) + \int_y \bar\eta(y)\Psi(y) + \int_x \bar\Psi(x)\eta(x) - i\int_x\int_z \bar\eta(z)\, S_F(z-x)\,\eta(x)$$

Let's rename the variables that are integrated over [won't change anything]

$$= \int_x\int_y \bar\Psi(x) i S_F^{-1}(x-y)\Psi(y) + \int_x\left[\bar\eta(x)\Psi(x) + \bar\Psi(x)\eta(x)\right] - i\int_x\int_y \bar\eta(x)\, S_F(x-y)\,\eta(y)$$

Now we put the result to the generating functional:

$$Z_0[\bar\eta,\eta] = \int D\Psi D\bar\Psi \, \exp\left[i\int_x\int_y \bar\Psi'(x) i S_F^{-1}(x-y)\Psi(y) + i\int_x\left[\bar\eta\Psi' + \bar\Psi'\eta\right]\right]$$

$$Z_0[\bar\eta,\eta] = \int D\Psi D\bar\Psi \, \exp\Bigg[i\Bigg(\int_x\int_y \bar\Psi(x) i S_F^{-1}(x-y)\Psi(y) + \underline{\int_x\left[\bar\eta\Psi + \bar\Psi\eta\right]} - \underline{i}\int_x\int_y \bar\eta(x) S_F(x-y)\eta(y)\Bigg)$$
$$+ i\Bigg(-\underline{\int_x\left[\bar\eta\Psi + \bar\Psi\eta\right]} + \underline{2i}\int_x\int_y \bar\eta(x) S_F(x-y)\eta(y)\Bigg)\Bigg]$$

We sum terms inside the exponentials and get

$$Z_0[\bar\eta,\eta] = \underbrace{\int D\Psi D\bar\Psi \, \exp\left[i\int_x\int_y \bar\Psi(x)\, i S_F^{-1}(x-y)\Psi(y)\right]}_{\text{this term is } Z_0[0,0]} \exp\left[i\int_x\int_y \bar\eta(x) i S_F(x-y)\eta(y)\right]$$

$$\Rightarrow \; Z_0[\bar\eta,\eta] = Z_0[0,0]\,\exp\left[-\int d^4x\, d^4y\, \bar\eta(x)\, S_F(x-y)\,\eta(y)\right] \quad \square$$

Exercise 2.7 The Yukawa theory is defined by the Lagrangian

$$\mathcal{L} = \mathcal{L}_{KG} + \mathcal{L}_D + g\phi\bar\Psi\Psi$$

The generating functional is $Z[J,\eta,\bar\eta] = \exp\left[-g\int_z \frac{\delta}{\delta J_z}\frac{\delta}{\delta \eta_z}\frac{\delta}{\delta\bar\eta_z}\right] Z_{KG}[J]\, Z_D[\eta,\bar\eta]$

$Z_{KG}[J] = \exp\left[\frac{i}{2}\int_x\int_y J(x) i\Delta_F(x-y) J(y)\right]$ and $Z_D[\eta,\bar\eta] = \exp\left[i\int_x\int_y \bar\eta(x) i S_F(x-y)\eta(y)\right]$

For $Z[0,0,0]$ all odd derivatives vanish (since we will have $J=0=\eta=0=\bar\eta$)

We expand the exponent and the first non-zero contribution comes from the second order term. In the second order we get

$$Z[J,\eta,\bar\eta] = \frac{(-g)^2}{2}\int_x\int_y \frac{\delta}{\delta J(x)}\frac{\delta}{\delta J(y)} Z_{KG}[J]\, \frac{\delta}{\delta \eta_a(x)}\frac{\delta}{\delta \bar\eta_a(x)}\frac{\delta}{\delta \eta_b(y)}\frac{\delta}{\delta \bar\eta_b(y)} Z_D[\eta,\bar\eta] + O(g^4)$$

Here the indices a and b are for the different spinors and repeated indices are summed over

$\Rightarrow \psi_a\bar\psi_a = \sum_{a=1}^{4}\psi_a\bar\psi_a$

Now to calculate $Z[0,0,0]$ we compute the functional derivative and then put the sources to be zero at the end.

Let's first calculate the Klein-Gordon part

We use the axioms: $\frac{\delta J(y)}{\delta J(x)} = \delta^{(4)}(x-y)$ and $\frac{\delta}{\delta J(x)}\int d^4y\, J(y)\phi(y) = \phi(x)$

$$\frac{\delta}{\delta J(x)} Z_{KG}[J] = \frac{\delta}{\delta J(x)} e^{\frac{i}{2}\int_k\int_l J(k)\, i\Delta_F(k-l)\, J(l)}$$

$$= \frac{i}{2}\left[\int_z i\Delta_F(x-z)J(z) + J(z)\, i\Delta_F(z-x)\right] Z_{KG}[J]$$

$$\Rightarrow \frac{\delta}{\delta J(x)} Z_{KG}[J] = \frac{i^2}{2}\int_z \left[\Delta_F(x-z)J(z) + J(z)\Delta_F(z-x)\right] Z_{KG}[J]$$

$$\Rightarrow \frac{\delta}{\delta J(x)}\frac{\delta}{\delta J(y)} Z_{KG}[J] = \frac{\delta}{\delta J(x)}\left[\frac{i^2}{2}\int_z\left(\Delta_F(y-z)J(z) + J(z)\Delta_F(z-y)\right) Z_{KG}[J]\right]$$

$$= \frac{i^2}{2}\left(\Delta_F(y-x) + \Delta_F(x-y)\right) Z_{KG}[J] + \text{Term that goes to 0 when } J=0$$

$$\Rightarrow \frac{\delta}{\delta J(x)}\frac{\delta}{J(y)} Z_{KG}[J]\Big|_{J=0} = \frac{i^2}{2}\left(\Delta_F(y-x) + \Delta_F(x-y)\right) = i^2\Delta_F(x-y) \qquad \Delta_F(x-y) = \Delta_F(y-x$$

Now we calculate the Dirac part, we have additiona functional derivative rule of

$$\frac{\delta \eta_b(y)}{\delta \eta_a(x)} = \delta(x-y)\,\delta_{ab}$$

$$\Rightarrow \frac{\delta}{\delta\bar\eta_b(y)} Z_D[\eta,\bar\eta] = \frac{\delta}{\delta\bar\eta_b(y)} e^{i\int_u\int_v \bar\eta(u)\, iS_F(u-v)\,\eta(v)} = \left[\frac{\delta}{\delta\bar\eta_b(y)}\, i\int_u\int_v \bar\eta_c(u)\, i\overbrace{S_F^{cd}}^{=S_F^{cd}(u-v)}\eta_d(v)\right] Z_D[\eta,\bar\eta]$$

$$\frac{\delta}{\delta\bar\eta_b(y)} Z_D[\eta,\bar\eta] = \left[i^2\int_v \delta_{cb}\, S_F^{cd}(y-v)\,\eta_d(v)\right] Z_D[\eta,\bar\eta] = \left(i^2\int_v S_F^{bd}(y-v)\,\eta_d(v)\right) Z_D[\eta,\bar\eta]$$

$$\Rightarrow \frac{\delta}{\delta\eta_b(y)}\frac{\delta}{\delta\bar\eta_b(y)} Z_D[\eta,\bar\eta] = i^2 S_F^{bb}(y-y)\, Z_D[\eta,\bar\eta] + i^2 i^2\int S_F^{bd}(y-v)\,\eta_d(v)\int_u \bar\eta_c(u)\, S_F^{cb}(u-y)\, Z_D[\eta,\bar\eta]$$

$$\frac{\delta}{\delta \bar{\eta}_a(x)} \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta,\bar{\eta}] = i^2 i^2 \left[S_F^{bb}(0) \int_v S_F^{ad}(x-v)\eta(v) \right] Z_D[\eta,\bar{\eta}] + i^2 i^2 \left[\int_v S_F^{bd}(y-v)\eta(v) \right] S_F^{ab}(x-y)\, Z_D[\eta,\bar{\eta}]$$

+ Higher order term that goes to 0 when $\eta = \bar{\eta} = 0$

$$\Rightarrow \frac{\delta}{\delta \eta_a(x)} \frac{\delta}{\delta \bar{\eta}_a(x)} \frac{\delta}{\delta \eta_b(y)} \frac{\delta}{\delta \bar{\eta}_b(y)} Z_D[\eta,\bar{\eta}] = S_F^{bb}(0)\, S_F^{aa}(0)\, Z_D[\eta,\bar{\eta}] + S_F^{ba}(y-x)\, S_F^{ab}(x-y)\, Z_D[\eta,\bar{\eta}] + 0$$

$$\Rightarrow Z[0,0,0] = \frac{(-g)^2}{2} \int_x \int_y \left[\frac{\delta}{\delta J(x)} \frac{\delta}{\delta J(y)} Z_{KG}[J] \frac{\delta\,\delta\,\delta\,\delta}{\delta \eta_a(x)\, \delta \bar{\eta}_a(x)\, \delta \eta_b(y)\, \delta \bar{\eta}_b(y)} Z_D[\eta,\bar{\eta}] \right] \Bigg|_{J=\eta=\bar{\eta}=0}$$

Now we substitute the results of derivatives

$$Z[0,0,0] = \frac{g^2}{2} \int_x \int_y i^2 \Delta_F(x-y) \underbrace{S_F^{bb}}_{\text{Trace}}(0) \underbrace{S_F^{aa}}_{\text{Trace}}(0)\, Z_D[0,0] + \underbrace{S_F^{ba}(y-x)\, S_F^{ab}(x-y)}_{\text{Trace}}\, Z_D[0,0]$$

$Z_D[0,0] =$

$$Z[0,0,0] = -\frac{g^2}{2} \int_x \int_y \Delta_F(x-y) \left[\mathrm{Tr}\left(S_F(y-x)\, S_F(x-y)\right) + \mathrm{Tr}\left(S_F(0)\right) \mathrm{Tr}\left(S_F(0)\right) \right]$$

The normalization factor $Z[0,0,0]$ to $O(g^2)$ is

$$\boxed{Z[0,0,0] = -\frac{g^2}{2} \int_x \int_y \Delta_F(x-y) \left[\mathrm{Tr}\left(S_F(y-x)\, S_F(x-y)\right) + \mathrm{Tr}\left(S_F(0)\right) \mathrm{Tr}\left(S_F(0)\right) \right]}$$